Resolution Geometry

Gravitation: The Field Equation Selected, Not Postulated

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Abstract

Clock curvature alone does not determine gravity, and the curvature of a declared-cost connection is not spacetime curvature. We close the passage to gravitation on Resolution Geometry's own terms. Einstein's field equation is not postulated. Each layer of the computation is a uniqueness theorem under a named principle, exactly as the Fisher metric is selected by Markov monotonicity:

1. light, through RG's cone stratum, fixes the conformal class;
2. anchored clocks fix the conformal factor;
3. conservation of the source fixes the motion of matter (Geroch–Jang);
4. RG's two-route consilience requires a local, metric-only, second-order, identically divergence-free relation between geometry and source, and in four dimensions this forces $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ (Lovelock);
5. spherical vacuum is then Schwarzschild without any staticity assumption (Birkhoff).

The selected equation, not a clock fit, is what excludes the clock-invisible γ-family: $G^r{}_r = 2m(\gamma - 1)/[r^2(2m - r)]$. Each principle has a countermodel, so none is decorative:

- Einstein's 1915 Ricci-only law is inconsistent for any star;
- $f(R)$ gravity is fourth order;
- the Gauss–Bonnet tensor survives only for $D \geq 5$;
- a pressure-blind source law is refuted by clock consilience.

What remains are two calibrations. The coupling $\kappa = 8\pi G$ is provably *only* a unit anchor, by RG's no-internal-unit theorem: exterior predictions in geometric units never contain G. The constant Λ is a window-certified constant.

Calibrated once from Newtonian Kepler orbits alone, the stack predicts four relativistic observations it never used:

- Mercury's perihelion advance, $42.98''$ per century;
- solar-limb light deflection, $1.751''$;
- the GPS satellite clock factory frequency, 10.22999999543 MHz;
- the Pound–Rebka shift, 2.46×10^{-15}.

The supplied computational backend is retained and reproduced byte-identically: vacuum and TOV equilibria, curvature, clocks, lensing, periapsis and radar, with 25 of 25 frozen synthetic checks and 6 of 6 symbolic identities. The new stack battery passes 16 of 16, and its four canaries are caught. Not claimed: a derivation from nothing, quantum gravity, rotation, radiation or cosmology.

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1 What was missing

The earlier draft computed gravity on two routes:

- *operational:* a static metric reconstructed from anchored clocks and null slopes, within a declared family;
- *dynamical:* Einstein's equation declared, and its spherical solutions computed.

Its ledger listed "Einstein dynamics from RG alone: not derived". That was correct for that draft. The missing steps were the layers that turn light, clocks and sources into a *forced* field equation. This paper supplies them, keeps the earlier obstructions, and changes the status of the field equation from *declared* to *selected*.

2 Obstructions retained

Theorem 2.1 (Task-level gravitational descent)**.** *For a nonempty observational fibre $\mathcal{F}_o$, a gravitational task H is identified iff it is constant on $\mathcal{F}_o$. For bounded data, the output is the task image $H(\mathcal{F}_D)$.*

No-Go 2.2 (Connection curvature is not Riemann curvature). The curvature $\Omega = (I - BJ)[Be_i, Be_j]$ is valued in $\ker DF$. Riemann curvature acts on TM. Moreover, no positive-definite cost becomes Lorentzian under an invertible congruence or a restriction. Gravity therefore cannot be read off the physical-curvature connection; it needs the causal route below.

No-Go 2.3 (Clock data alone do not determine gravity). In the family $A = 1 - 2m/r$, $C_\gamma = (1 - 2\gamma m/r)^{-1}$, every γ gives the same static clock rates, but different light bending: at $r_0 = 1000m$ the deflections are 0.002004, 0.004008 and 0.006016 for $\gamma = 0$, 1, 2. A conformal factor equal to 1 at the clock stations also changes massive geodesics while keeping null curves.

3 The gravity stack

3.1 Light fixes the conformal class

Proposition 3.1 (Null orbits are conformal-class data; GS06)**.** *For $-A\,\mathrm{d}t^2 + C\,\mathrm{d}r^2 + D\,\mathrm{d}\Omega^2$, the null orbit equation $\mathrm{d}\phi/\mathrm{d}r = \sqrt{C}\,(b/D)/\sqrt{1/A - b^2/D}$ is invariant under $(A, C, D) \to \Omega^2(A, C, D)$. The deflection is identical for g and $\Omega^2 g$ (0.5002356566 rad at $r_0 = 10m$), while clock rates differ by Ω.*

RG's cone stratum (ladder L64–L68) reconstructs the conformal class from causal data. Here, null orbits supply it [5, 6, 7].

3.2 Anchored clocks fix the scale

Theorem 3.2 (Static clock–radar reconstruction)**.** *In an anchored areal chart, a static clock rate $N = \sqrt{A}$ and a radial null slope $h = \sqrt{C/A}$ give $A = N^2$ and $C = N^2h^2$ uniquely. Locally, ten anchored timelike proper-rate directions determine all ten metric coefficients (rank 10, condition number 176.67, error 8.9×10^{-16}), while repetitions of a single static clock direction give rank 1.*

The anchor is RG's counted channel (L94–L109). Clocks plus null data, unlike clocks alone, fix g.

3.3 Conservation fixes motion

Proposition 3.3 (Motion from $\nabla \cdot T = 0$; GS05)**.** *For a static perfect fluid, $\nabla_\mu T^{\mu r} = 0$ is exactly $p' = -(\rho + p)\nu'$: hydrostatic support, in which pressure gravitates. For pressureless matter it demands $\nu' = 0$, so dust cannot hover; it falls freely.*

Imported Theorem 3.4 (Geroch–Jang [3])**.** *A small body with conserved stress–energy satisfying the dominant energy condition follows a timelike geodesic.*

Geodesic motion is therefore derived from conservation, not postulated.

3.4 Consilience selects the field equation

RG's two routes to the source, the geometric route and the source route (*The Stress Tensor*), are related by a law $E[g] = \kappa T$. The following principles are declared, and each is tested by a countermodel:

Selection principle 3.5 (P1 locality, P2 metric only, P3 second order, P4 compatibility, P5 dimension)**.**

P1 E is built locally from g at a point.

P2 E uses no gravitational field besides g.

P3 E uses at most second derivatives of g.

P4 E is symmetric and identically divergence-free, since $\nabla \cdot T = 0$ for *every* source.

P5 The dimension is $D = 4$, which is the local rank of anchored clock tomography.

Imported Theorem 3.6 (Lovelock [2])**.** *Under P1–P5, $E_{\mu\nu} = aG_{\mu\nu} + bg_{\mu\nu}$.*

Proposition 3.7 (The Einstein tensor is forced; GS01)**.** *For every static spherical metric, $\nabla^\mu(aR_{\mu\nu} + bRg_{\mu\nu}) = (a/2 + b)\,\partial_\nu R$ exactly. Identical conservation forces $b = -a/2$. A canary that accepts Ricci alone ($b = 0$) is caught.*

Countermodels: no principle is decorative.

- *P4.* Einstein's 1915 trial $R_{\mu\nu} = \kappa T_{\mu\nu}$ would force the trace of T to be constant. For a polytropic star the trace varies by more than 100%, so the law is INCONSISTENT (GS02).
- *P3.* The $f(R) = R + \alpha R^2$ field equation contains fourth derivatives of the metric (GS04).
- *P5.* The Gauss–Bonnet tensor, second order and identically divergence-free, vanishes identically in $D = 4$ but not in $D = 5$ (GS03).
- *P2.* Scalar–tensor theories add a field and change γ, which the held-out light deflection tests (RW02).
- *P1.* Nonlocal modifications are excluded by declaration only.

3.5 Vacuum: Birkhoff, and the end of the γ-family

Proposition 3.8 (Birkhoff; GS07)**.** *For $-e^{2a(t,r)}\mathrm{d}t^2 + e^{2b(t,r)}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2$, the vacuum component $G_{tr} = 2\,\partial_t b/r$ forces b to be static. The remaining equations give Schwarzschild, and a's time dependence is absorbed into t. The exterior of a pulsating or collapsing star is static by theorem.*

Proposition 3.9 (The selected equation excludes $\gamma \neq 1$; GS08)**.** *In the γ-family, $G^t{}_t \equiv 0$ for every γ, but*

$$G^r{}_r = \frac{2m(\gamma - 1)}{r^2(2m - r)}.$$

The rr equation, which couples the clock function to the radial geometry, is what excludes the clock-invisible alternatives.

3.6 Calibrations: what no theory can derive

Proposition 3.10 (κ is a unit anchor; GS09)**.** *In geometric units, every exterior task depends only on the length $m = GM/c^2$, which clocks measure, so G never appears. Interior computations have no internal scale. Under the TOV homology $(r, m) \to \lambda(r, m)$, $(\rho, p) \to (\rho, p)/\lambda^2$, with the equation of state rescaled, a star maps to a star (ratios* 3.0000000*). Converting kilograms to gravitational length therefore needs $\kappa = 8\pi G/c^4$ as a calibration. By RG's no-internal-unit theorem this is* all *it can be.*

Proposition 3.11 (Λ is a window; GS11)**.** *With clock error ϵ at radius $R_{\max}$, the clocks give $|\Lambda| \leq 3\epsilon/R_{\max}^2$. This is a resolution-indexed certificate (L114): Λ is LOCKED within a window, or declared by an asymptotic boundary.*

4 Computation

The retained backend (`rg_gravity.py`) computes the following:

- *vacuum:* $m' = 0$, $A = 1 - 2m/r$, $C = A^{-1}$, with ODE agreement 3.9×10^{-12};
- *fluid:* the coupled TOV system

$$m' = 4\pi r^2 \rho, \qquad \nu' = \frac{m + 4\pi r^3 p}{r(r - 2m)}, \qquad p' = -(\rho + p)\nu'.$$

  The constant-density interior is exact to 10^{-12}. A $\Gamma = 2$ polytrope ($K = 100$, $\rho_{0c} = 10^{-3}$) gives $R = 10.04735$ and $M = 1.26957$. A retained surface discrepancy of 6.4×10^{-5} between solvers is reported, not hidden.

- *curvature:* $R_{\mu\nu} = 0$ and $K = 48m^2/r^6$, with tidal eigenvalues $(m/r^3)(2, -1, -1)$;

- *observables:*
  - static redshift $\sqrt{A(r_e)/A(r_r)}$;
  - orbiting clock $\mathrm{d}\tau/\mathrm{d}t = \sqrt{1-3m/r}$;
  - exact deflection, against the null orbit $u'' + u = 3mu^2$;
  - periapsis advance;
  - radial radar $2(r_2 - r_1) + 4m\log[(r_2 - 2m)/(r_1 - 2m)]$;
- *Newtonian limit:* the potential, with second-order grid convergence.

Proposition 4.1 (Source–clock consilience; GS10)**.** *For the polytrope, the source route through the selected equation gives a mass equal to the exterior clock mass (*1.269574*). The same star under a pressure-blind Newtonian source law gives* 2.193 *and is INCONSISTENT with the clocks. Across central densities, the selected equation shows a maximum mass, because pressure gravitates; the Newtonian law has none. Two interiors with the same exterior mass share the exterior, so the interior remains a CORRESPONDENCE.*

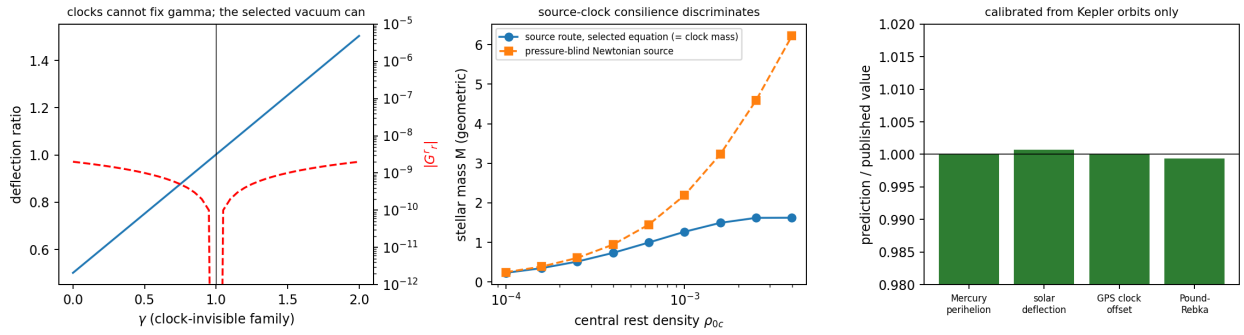


Figure 1: Left: the clock-invisible γ-family. Deflection grows linearly with γ, while the selected vacuum residual $|G^r{}_r|$ vanishes only at $\gamma = 1$. Centre: stellar mass from the source route under the selected equation, which equals the clock mass and saturates at a maximum, against a pressure-blind Newtonian source law. Right: four relativistic predictions from Newtonian Kepler calibrations, divided by their published values.

5 Calibrate once, predict the relativistic

Only Newtonian orbital data are used for calibration. Kepler's law $GM = 4\pi^2 a^3/T^2$ gives:

- from Earth's year and the astronomical unit, $GM_\odot = 1.32713 \times 10^{20}\,\mathrm{m^3\,s^{-2}}$;
- from the GPS orbit ($a = 26561.75\,\mathrm{km}$, $T = 43082.05\,\mathrm{s}$), $GM_\oplus = 3.98600 \times 10^{14}\,\mathrm{m^3\,s^{-2}}$.

No relativistic observation enters.

prediction	formula (selected equation)	RG value	published
Mercury perihelion (per century)	$6\pi GM/[c^2 a(1-e^2)]$	$42.98''$	$\approx 43''$ anomaly [8]
solar-limb deflection	$4GM/(c^2 R_\odot)$, with $\gamma = 1$ forced	$1.751''$	confirmed, VLBI [8]
GPS clock factory frequency	$L_G - 3GM_\oplus/(2c^2 a)$	10.22999999543 MHz	as built [9]
Pound–Rebka shift (22.5 m)	gh/c^2	2.46×10^{-15}	percent agreement [10]

Here $L_G = 6.969290134 \times 10^{-10}$ is the IAU geoid clock-rate constant, the anchor of the ground clock. The predictions test different layers of the stack:

- the two clock predictions test the clock and scale layer (equivalence and the weak field);
- the deflection tests the conformal class and the vacuum selection of γ;
- the perihelion advance tests the nonlinear vacuum.

The published values are literature anchors used for comparison, not new data. Canaries with the Newtonian corpuscular deflection (half the value) and with no relativistic advance are caught.

6 Frozen synthetic holdout (retained)

Separate processes calibrate, predict and score twelve held-out cases with six GR tasks each, plus an operational family with two tasks per case. A rerun regenerates byte-identical predictions.

Check	Result
Final gravity battery	25/25 checks passed
Separate symbolic audit	6/6 identities; included once in the battery wrapper
Frozen GR task scores	12 cases, six observables each; all passed
Frozen operational-family scores	12 cases, two observables each; all passed
Maximum relative deflection error	3.453×10^{-9}
Maximum relative precession error	3.897×10^{-9}
Maximum relative static frequency-ratio error	2.706×10^{-10}
Maximum relative radial-excess error	2.810×10^{-9}
Maximum relative proper-radar error	1.167×10^{-10}
Maximum relative orbiting-clock error	2.107×10^{-10}
Maximum relative operational-family error	3.710×10^{-8}
Conditional GR task intervals	48/48 contain the synthetic outcomes within declared solver slack
Largest relative interval diameter	2.416×10^{-7}
Largest geodesic conservation residual	1.757×10^{-12}
Largest ODE-refinement task change	1.251×10^{-10}
Mutations and negative controls	12/12 detected; four output mutations reject all 12 cases each
Invalid-domain inputs	8/8 rejected

Limits. The freeze is computational, not external blinding. Predictor and oracle share the declared metric. All records are synthetic. Tolerances are engineering gates, not interval-arithmetic bounds.

7 Ledger

layer	status
conformal class	LOCKED from null data (cone stratum; conformal invariance)
metric scale	LOCKED from anchored clocks; local rank 10
motion of matter	theorem from conservation (Geroch–Jang); not postulated

layer	status
field equation	SELECTED: $G + \Lambda g = \kappa T$ under P1–P5 (Lovelock). Einstein's postulate is replaced by a selection theorem
spherical vacuum	Schwarzschild by Birkhoff; $\gamma = 1$ forced by the rr equation
$\kappa = 8\pi G$	unit calibration; provably not derivable (homology, no internal unit)
Λ	calibrated window, or boundary declaration
interior source from exterior	CORRESPONDENCE
imported	mathematical theorems (Lovelock, Geroch–Jang, Birkhoff, conformal invariance of null geodesics); principles P1–P5
not claimed	derivation from nothing; quantum gravity; rotation, radiation, cosmology, mergers; laboratory validation of P1–P5

8 What this is and is not

It is gravity computed on RG's terms:

- every layer is a uniqueness theorem under a named principle, as with Fisher (Čencov);
- every principle has a countermodel;
- the only constants are calibrations, one of which (the unit anchor) no theory can derive;
- calibrations from one regime (Newtonian orbits) predict another (relativistic clocks, light and orbits).

It is not a derivation of gravity from equality of laws alone. P1–P5 are physical selection principles. Their failure is testable, and the countermodels show what failure would look like. No priority is claimed: the selection theorems are classical [2, 4, 3, 5].

A Reproducibility

- `rg_gravity_battery.py` (retained) passes 25/25, together with the symbolic audit's 6/6. The rerun is byte-identical, and `score` mode rescores without regeneration.
- `rg_gravity_stack_attack.py` (GS01–GS12, RW01–RW04) passes 16/16. Its four canaries are caught: a pressure-blind source, Ricci alone, Newtonian deflection, and no advance.

Two corrections were made before registration:

1. GS08 first tested $G^t{}_t$, which vanishes for every γ; the rr component carries the selection.
2. GS09's homology test first used fixed surface thresholds and step sizes, which broke the symmetry numerically.

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